Operating Systems - Quick Interview Notes

Processes, Threads, Memory, Scheduling

1. Process vs Thread

- Process: independent program in execution with its own memory space.
- Thread: lightweight unit of execution inside a process, shares memory.
- Context switch between threads is cheaper than between processes.

2. Process States

- New -> Ready -> Running -> Blocked/Waiting -> Terminated.
- Only one process runs on a single CPU core at a time.

3. CPU Scheduling Algorithms

- FCFS: First Come First Served (simple, convoy effect).
- SJF: Shortest Job First (min avg waiting, needs burst time).
- Round Robin: time slices, good for interactive systems.
- Priority Scheduling: can starve low-priority processes.
- MLFQ: Multiple Level Feedback Queues (used in modern OS).

4. Deadlock - 4 Necessary Conditions

- Mutual Exclusion, Hold and Wait, No Preemption, Circular Wait.
- Avoid via Banker Algorithm (safe state), prevent by breaking one condition.

5. Memory Management

- Paging: fixed-size pages/frames, no external fragmentation.
- Segmentation: variable-size segments, external fragmentation possible.
- Page replacement: FIFO, LRU, Optimal, Clock (Second Chance).
- Thrashing: excessive paging due to too many active pages.

6. Virtual Memory

- Allows programs larger than physical RAM using demand paging.
- TLB (Translation Lookaside Buffer) caches page-table entries.

7. Inter-Process Communication (IPC)

- Shared Memory and Message Passing.
- Pipes (unidirectional), message queues, semaphores, sockets.

8. Common Interview Topics

- Semaphores vs Mutex: counting vs binary lock.
- Starvation, Race condition, Critical Section problem.
- Zombie (child done, not reaped) vs Orphan (parent died) process.